

IMPACT OF AI-DRIVEN DEMAND FORECASTING AND INVENTORY OPTIMIZATION ON SUPPLY CHAIN PERFORMANCE

Course: Project Title
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Introduction & Background

- Supply chain management is vital for business efficiency and competitiveness.
- Traditional forecasting methods struggle with changing market conditions.
- AI technologies improve forecasting accuracy and inventory decisions.
- AI enables data-driven and real-time supply chain operations.

Problem Statement

- Conventional forecasting methods face limitations in dynamic markets.
- Businesses experience overstocking, understocking, and higher costs.
- Organizations require smarter forecasting and inventory systems.

Objectives of the Study

- To study the role of AI in demand forecasting.
- To examine AI-based inventory optimization.
- To analyze the relationship between forecasting and inventory management.
- To evaluate the impact of AI on supply chain performance.

Research Methodology

- Research Type: Descriptive and analytical study.
- Data Collection Method: Structured questionnaire.
- Respondents: Supply chain and logistics professionals.
- Statistical Tools: Descriptive analysis, correlation, regression, and ANOVA.

Literature Review

1. Aghaei et al. (2025) – AI improves supply chain efficiency and visibility.
2. Khan (2024) – AI & IoT enable real-time monitoring and responsiveness.
3. Verma (2024) – AI enhances forecasting accuracy.
4. Oviedo-Bayas et al. (2025) – ML reduces demand uncertainty.
5. Zhang et al. (2024) – Deep learning improves demand prediction

Literature review

6. Singh & Patel (2024) – AI optimizes inventory management.
7. Dunka (2024) – Predictive analytics reduces excess inventory.
8. White (2024) – ML determines optimal stock levels.
9. Kagalwala et al. (2025) – AI provides insights for logistics decisions.
10. Douaioui et al. (2024) – ML/DL improve supply chain planning

Literature Review

11. Pattnaik et al. (2024) – AI enhances operational efficiency.
12. Teixeira et al. (2025) – AI strengthens resilience and sustainability.
13. Samuels (2025) – AI supports intelligent supply chains.
14. Jauhar et al. (2025) – Explainable AI improves transparency.
15. Fatima & Salam (2026) – XGBoost improves inventory forecasting.

Literature Review

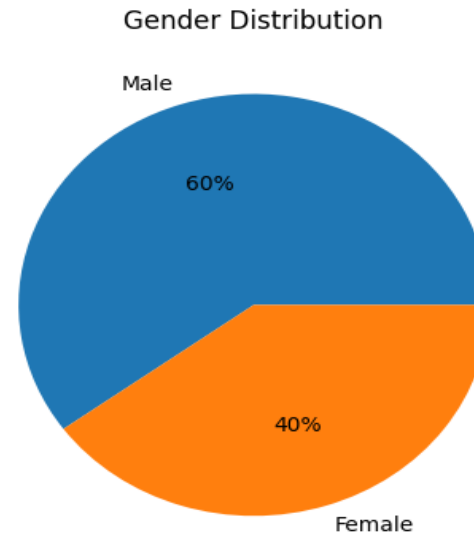
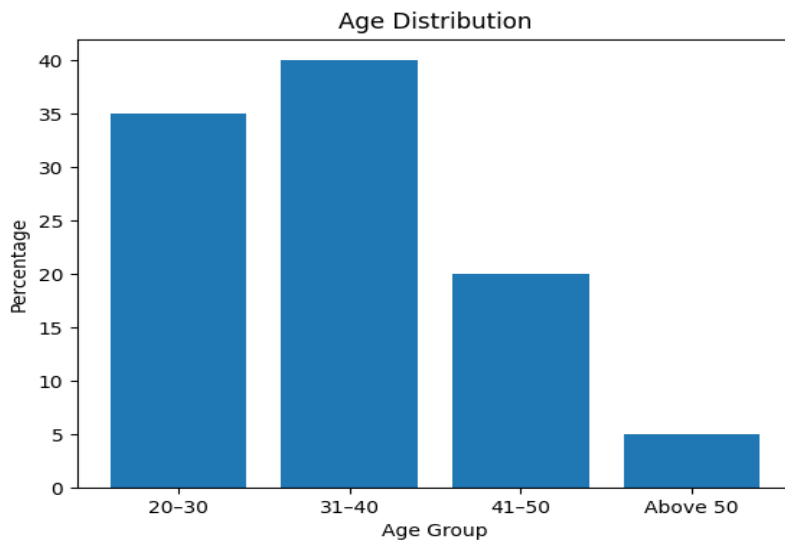
16. Wang et al. (2025) – ML increases inventory turnover.
17. Shahnawaz & Safder (2025) – AI reduces operational costs.
18. Yoshizato et al. (2026) – AI agents improve ordering decisions.
19. MDPI Survey (2025) – AI enhances end-to-end supply chain functions.
20. Reuters Report (2026) – AI improves reliability and efficiency.

Research Gap

- Most studies focus on forecasting or inventory separately.
- Limited empirical research on combined AI impact.
- Need for analysis on efficiency, cost savings, and customer satisfaction.

Demographic Profile

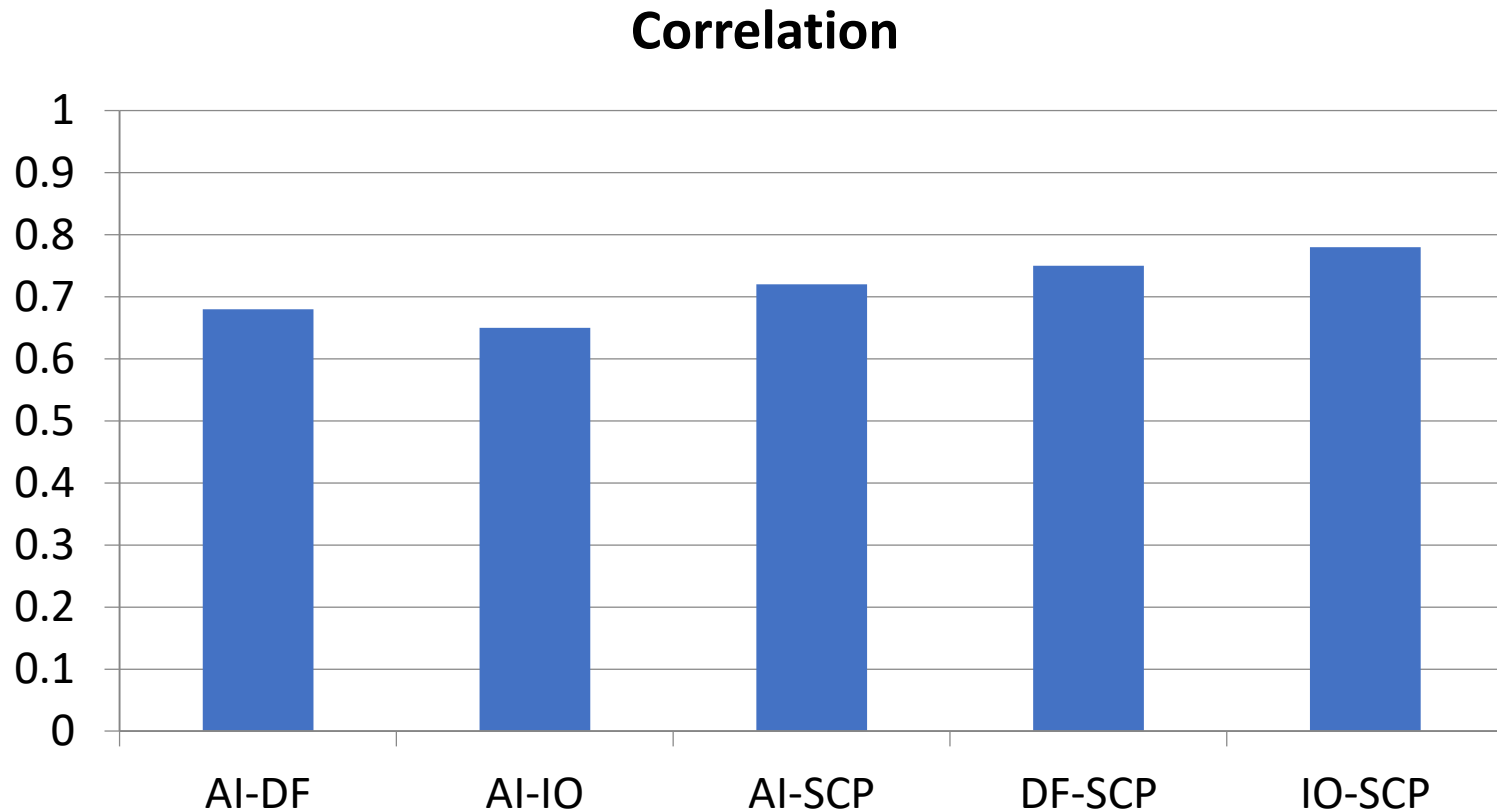
- 60% male respondents and 40% female respondents.
- Majority respondents were aged 31–40 years.
- 45% respondents had 5–10 years of experience.



Descriptive Analysis

- AI improves decision-making and minimizes human error.
- AI-based forecasting reduces demand uncertainty.
- AI enhances inventory planning and reduces stock issues.

Correlation Analysis

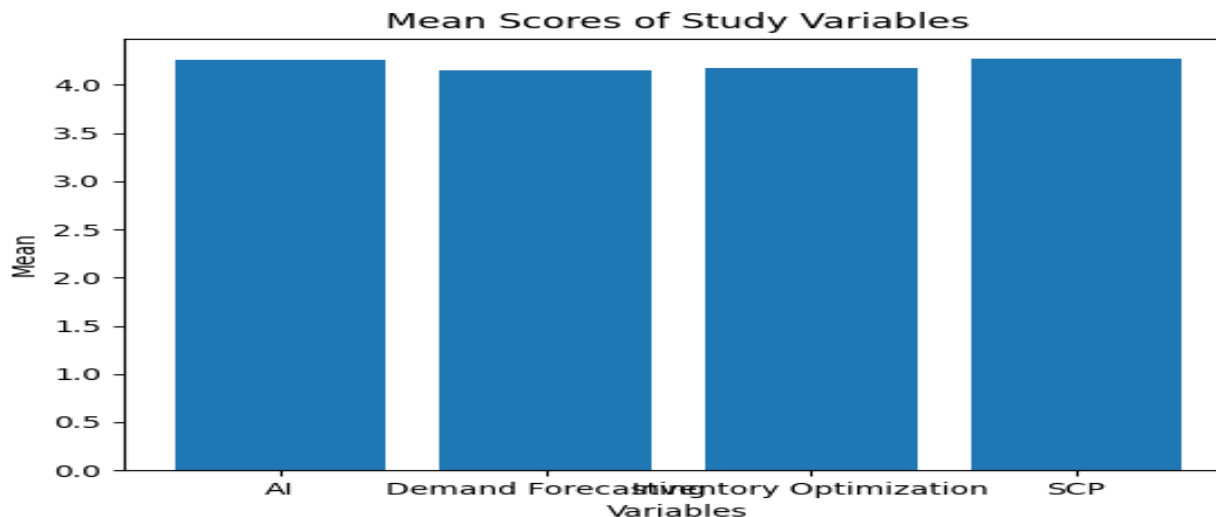


Regression Analysis

Variable	Beta	Significance
AI → SCP	0.62	0.000
AI → DF	0.68	0.000
DF → SCP	0.55	0.000
AI → IO	0.65	0.000
IO → SCP	0.60	0.000

Key Findings

- AI significantly improves demand forecasting accuracy.
- AI optimizes inventory through real-time monitoring.
- Positive relationship exists between forecasting and inventory management.
- AI improves supply chain efficiency and customer satisfaction.



Recommendations

- Implement AI-based forecasting systems.
- Use real-time inventory monitoring solutions.
- Train employees in AI and analytics tools.
- Develop integrated digital supply chain systems.

Conclusion

- AI is transforming supply chain management.
- AI improves forecasting, inventory optimization, and efficiency.
- AI adoption provides competitive advantage and cost savings.
- Future supply chains will increasingly rely on AI technologies.

Thank You

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